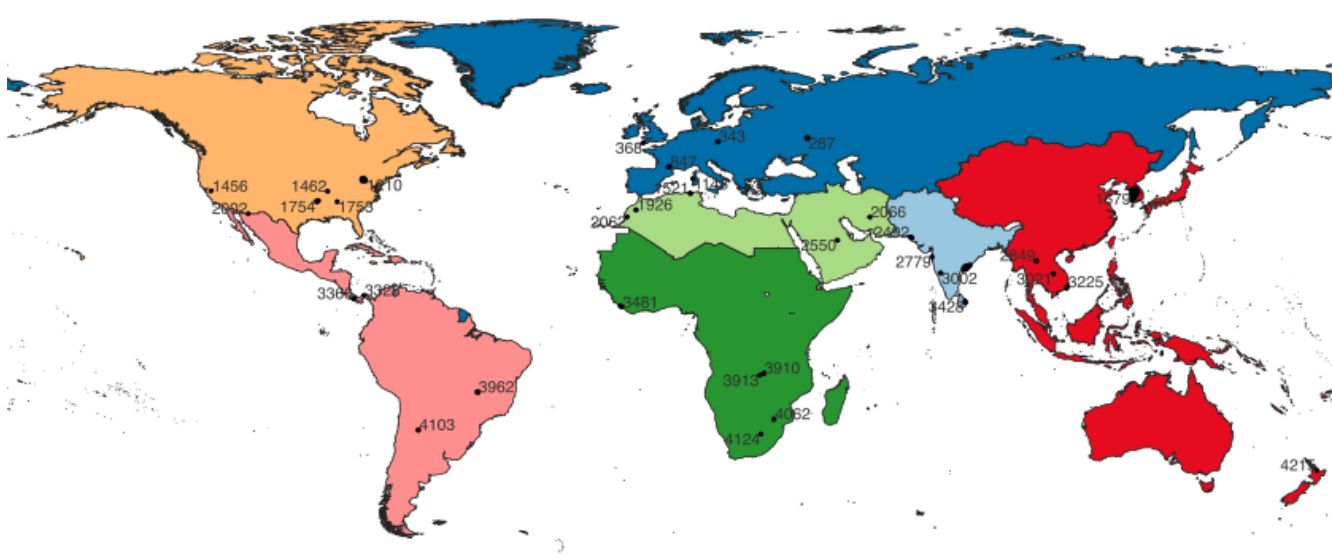


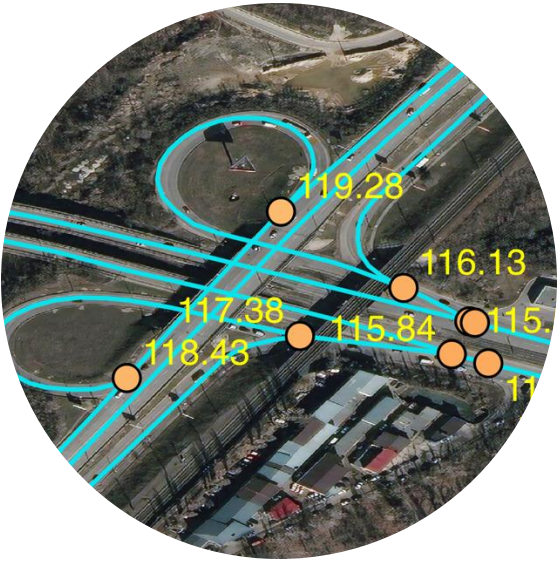
# Road network elevation consideration

This document provides details about the consideration of the elevation of road segments in our project. Road network datasets were extracted from OpenStreetMap, and elevation values were obtained using the Google Maps Elevation API. The API returns elevation in meters relative to the local mean sea level (LMSL), which we assigned to the nodes of each road network model.

To ensure that elevated road structures were accurately represented, we randomly sampled five networks from each of seven world regions and verified that these structures were correctly captured (**Figure 1**). In total, 35 networks distributed across 26 countries and regions were analyzed. The following pages present examples of selected networks that contain elevated road structures, along with the corresponding node elevation values.

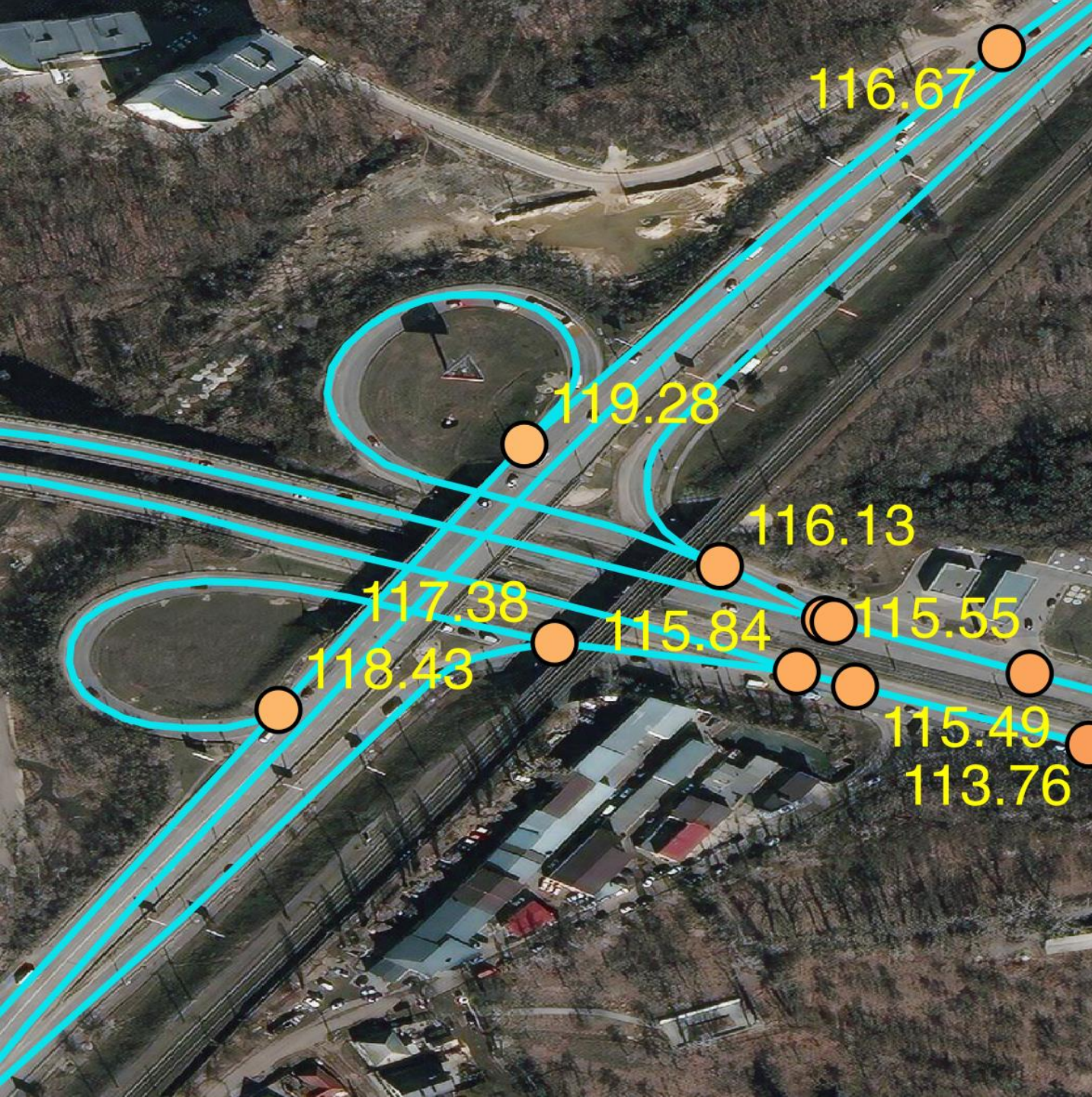


**Figure 1.** Sampled road networks in seven world regions. For each world region, we randomly sampled five settlement clusters and examine the accuracy of road segment elevation represented in the network model.



**Figure 2.** Representation of elevated road structures in the network model. Nodes are depicted as orange points, and road segments (edges) are shown as neon blue lines. Node elevations, expressed in meters relative to local mean sea level, are annotated in yellow. The basemap imagery is sourced from Bing Satellite.

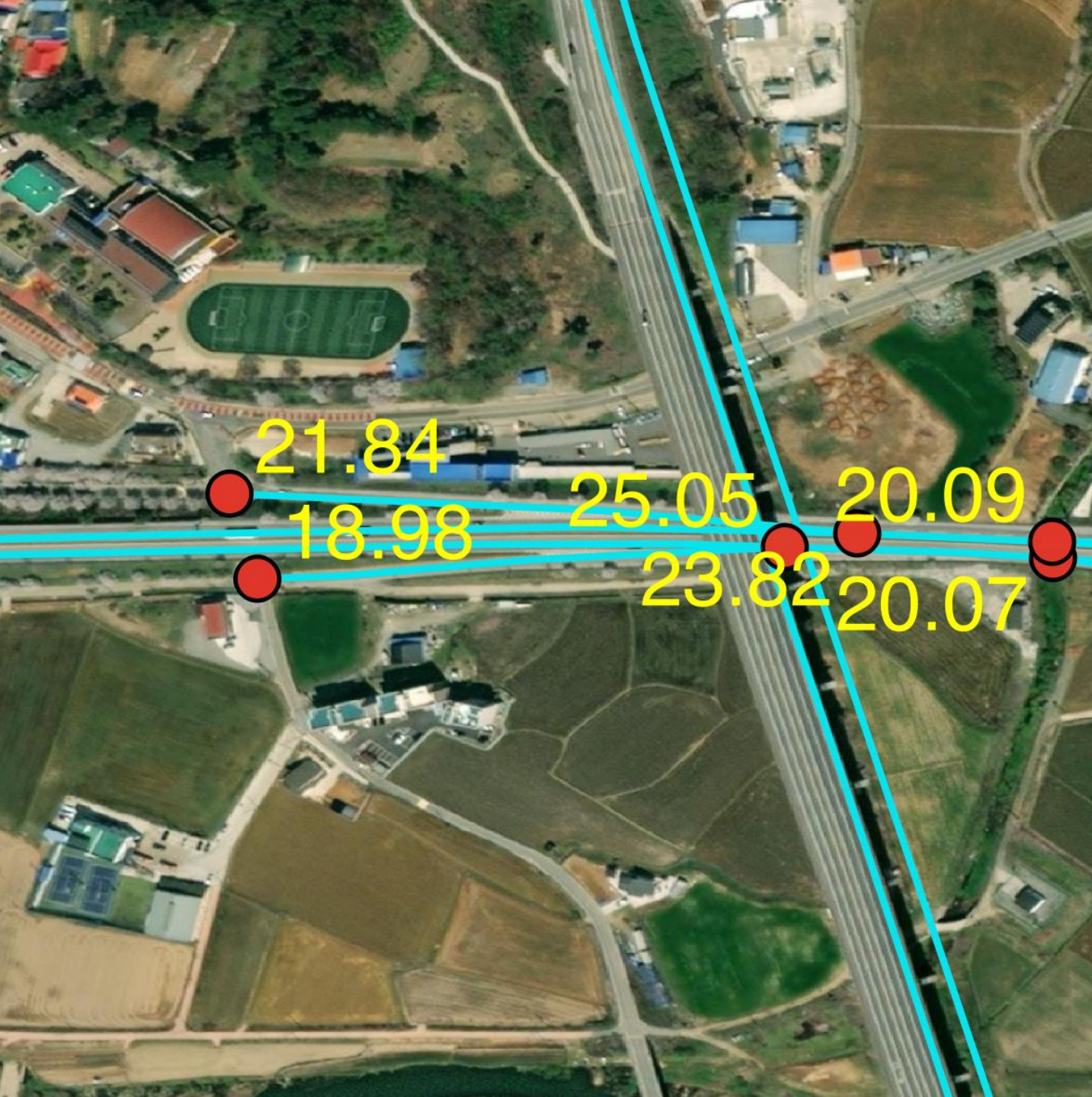




|                         |                       |
|-------------------------|-----------------------|
| Network ID              | 287                   |
| Closest city            | Voronezh              |
| Country                 | Russia                |
| World Region            | Europe & Central Asia |
| Elevated road structure | Overpass              |
| Difference in elevation | 3.73 meters           |







|                         |                     |
|-------------------------|---------------------|
| Network ID              | 1679                |
| Closest city            | Seoul               |
| Country                 | Korea               |
| World Region            | East Asia & Pacific |
| Elevated road structure | Overpass            |
| Difference in elevation | 4.96 meters         |



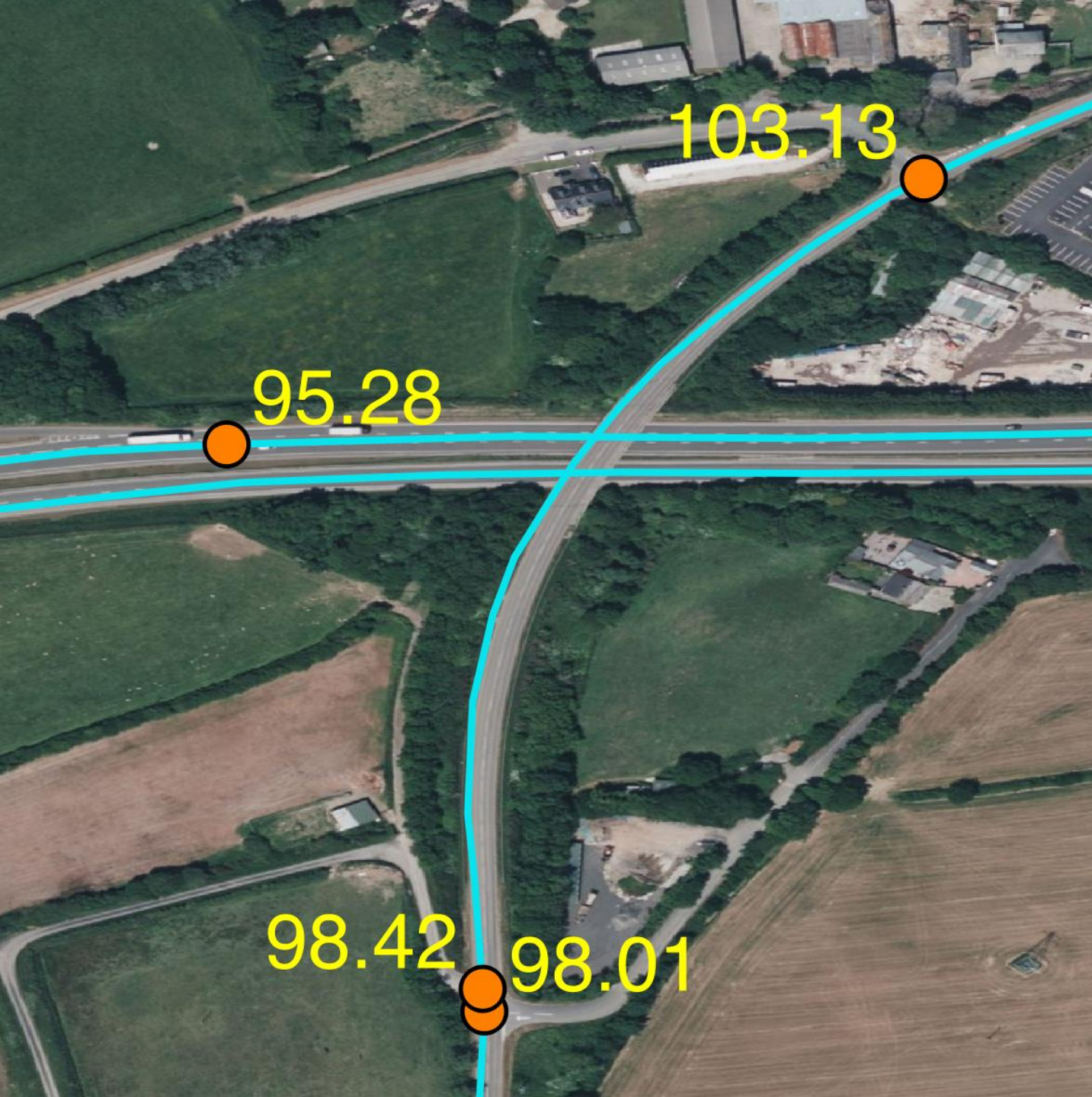




|                         |                       |
|-------------------------|-----------------------|
| Network ID              | 1143                  |
| Closest city            | Sassari               |
| Country                 | Italy                 |
| World Region            | Europe & Central Asia |
| Elevated road structure | Overpass              |
| Difference in elevation | 3.03 meters           |



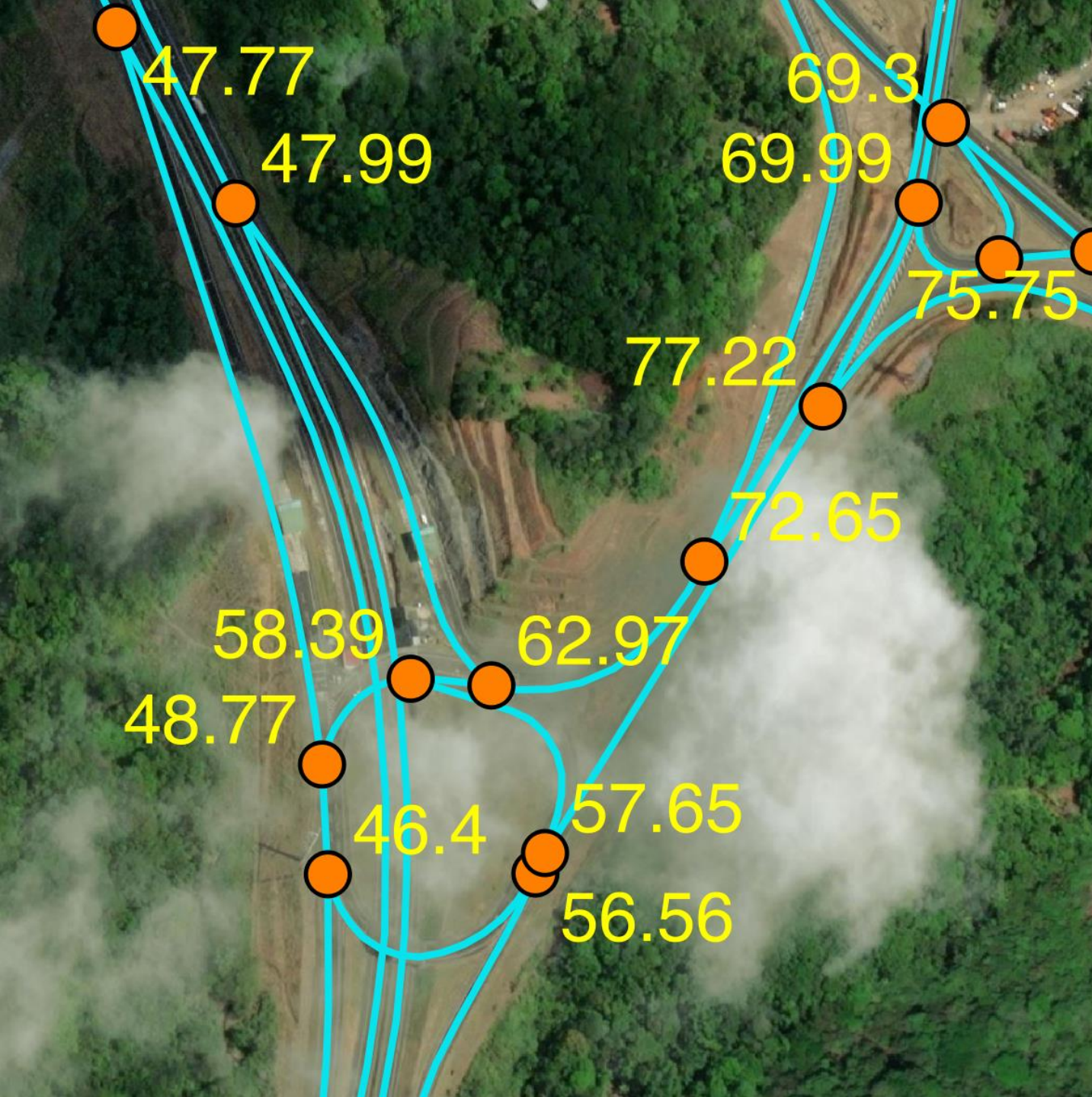




|                         |                       |
|-------------------------|-----------------------|
| Network ID              | 368                   |
| Closest city            | St Austell            |
| Country                 | United Kingdom        |
| World Region            | Europe & Central Asia |
| Elevated road structure | Overpass              |
| Difference in elevation | 7.85 meters           |



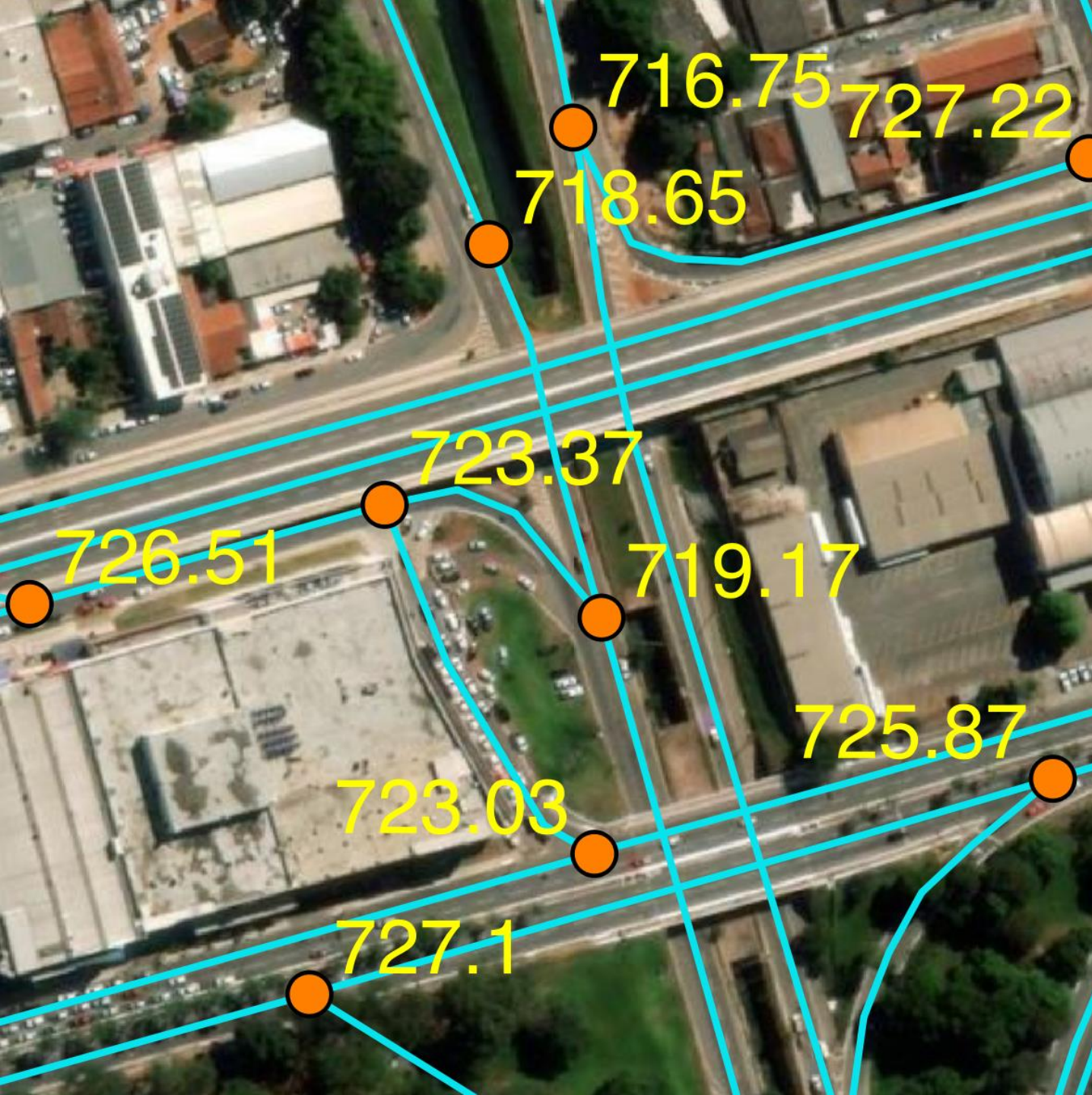




|                         |                           |
|-------------------------|---------------------------|
| Network ID              | 3328                      |
| Closest city            | Colón                     |
| Country                 | Panama                    |
| World Region            | Latin America & Caribbean |
| Elevated road structure | Elevated roadway          |
| Difference in elevation | 10.4 meters               |



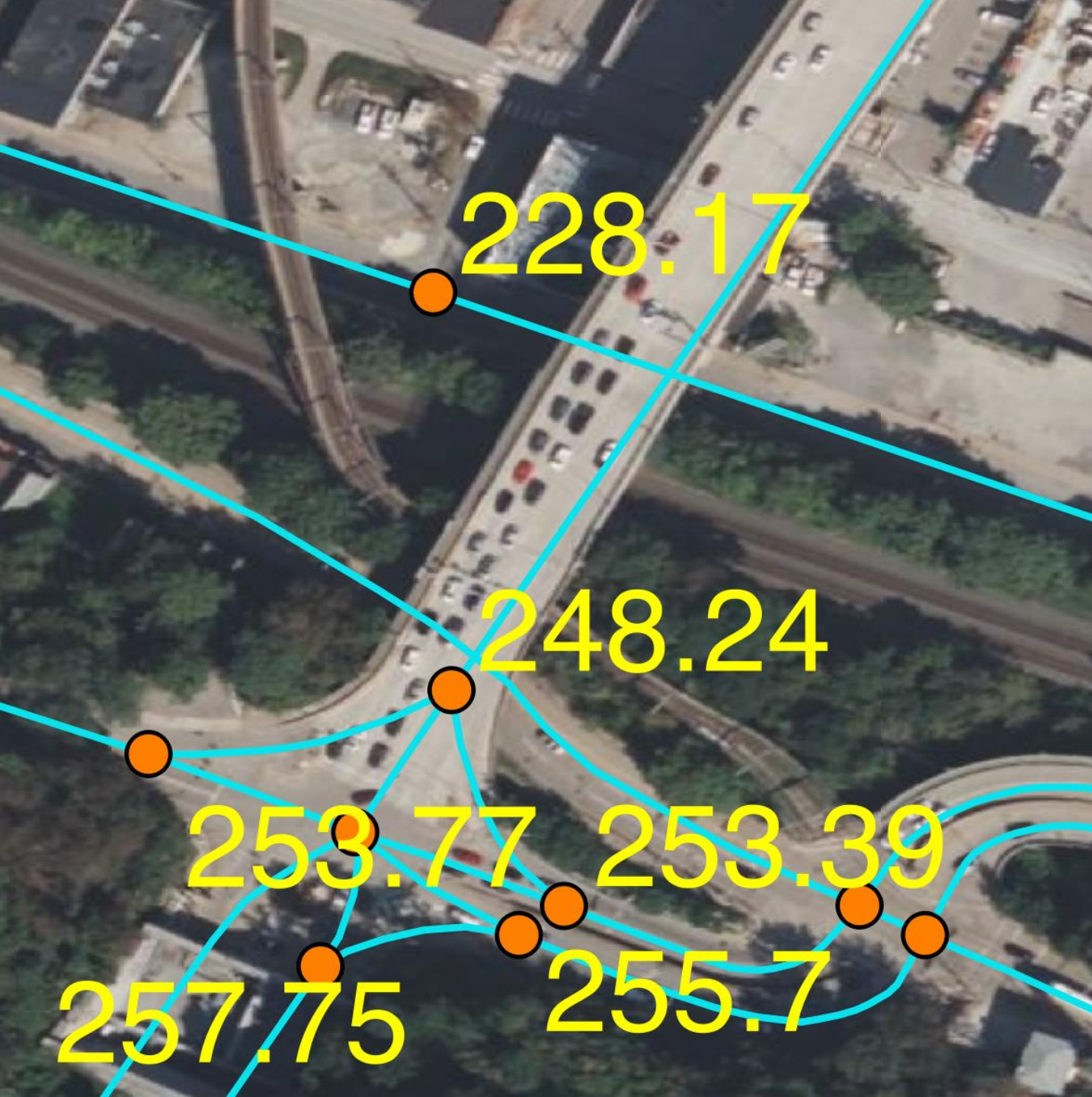




|                         |                           |
|-------------------------|---------------------------|
| Network ID              | 3962                      |
| Closest city            | St. Leste Vila Nova       |
| Country                 | Brazil                    |
| World Region            | Latin America & Caribbean |
| Elevated road structure | Overpass                  |
| Difference in elevation | 8.44 meters               |







|                         |               |
|-------------------------|---------------|
| Network ID              | 1210          |
| Closest city            | Pittsburgh    |
| Country                 | United States |
| World Region            | North America |
| Elevated road structure | Overpass      |
| Difference in elevation | 20.07 meters  |



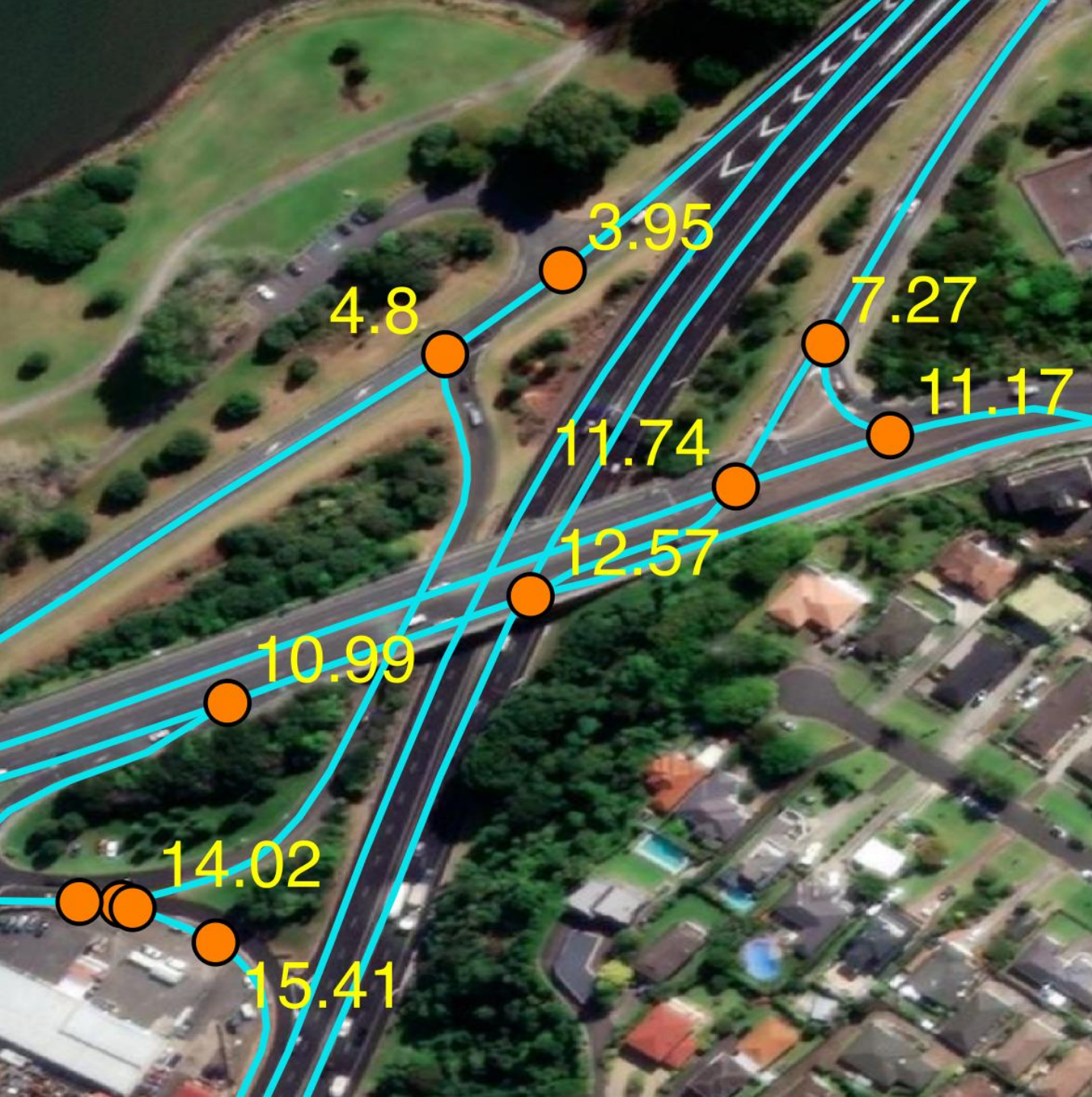




|                         |                     |
|-------------------------|---------------------|
| Network ID              | 1679                |
| Closest city            | Seoul               |
| Country                 | Korea               |
| World Region            | East Asia & Pacific |
| Elevated road structure | Bridge              |
| Difference in elevation | 5.55 meters         |



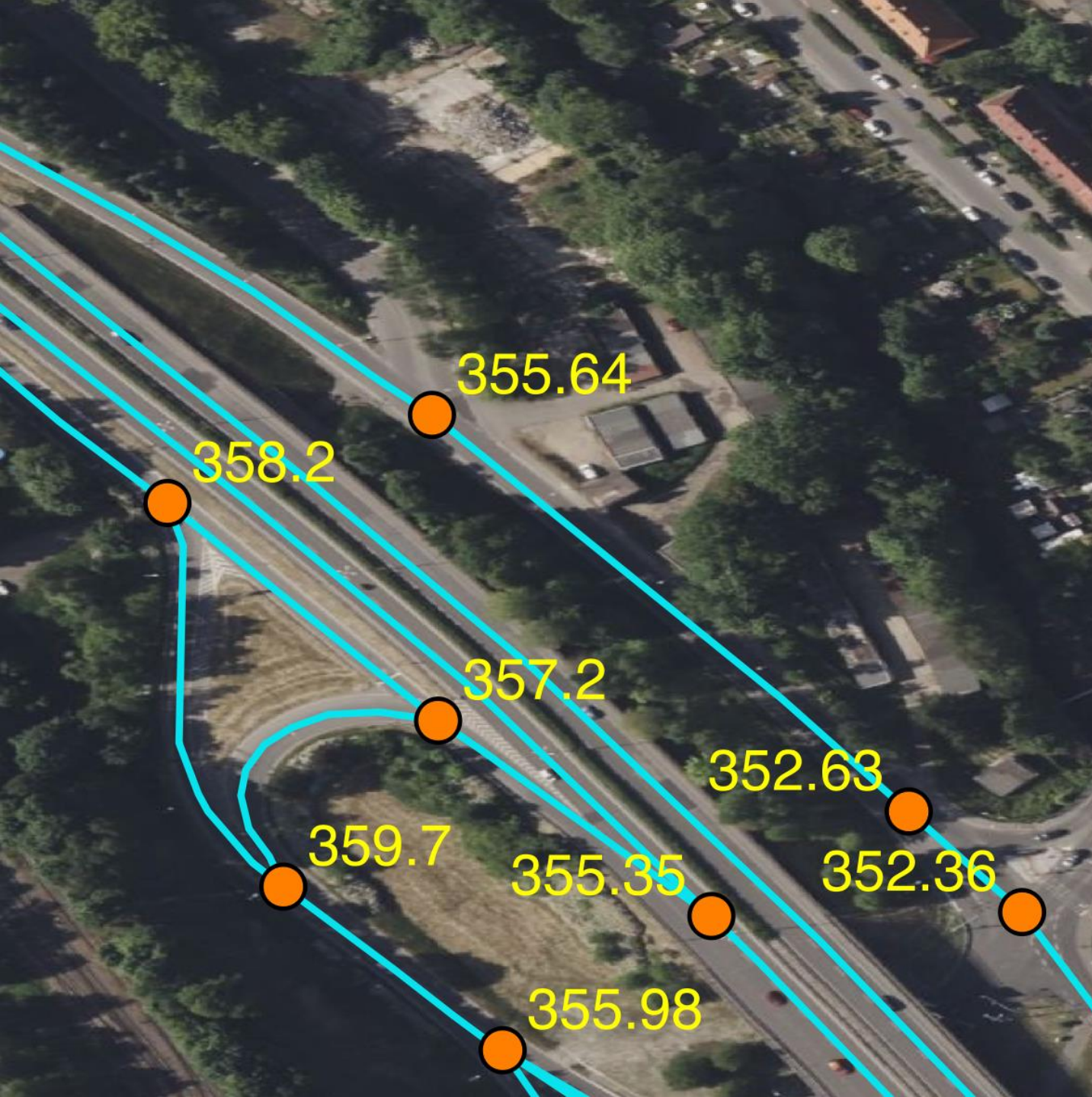




|                         |                     |
|-------------------------|---------------------|
| Network ID              | 4215                |
| Closest city            | Tauranga            |
| Country                 | New Zealand         |
| World Region            | East Asia & Pacific |
| Elevated road structure | Overpass            |
| Difference in elevation | 7.77 meters         |







|                         |                       |
|-------------------------|-----------------------|
| Network ID              | 343                   |
| Closest city            | Jablonec nad Nisou    |
| Country                 | Czechia               |
| World Region            | Europe & Central Asia |
| Elevated road structure | Elevated expressway   |
| Difference in elevation | 5.84 meters           |



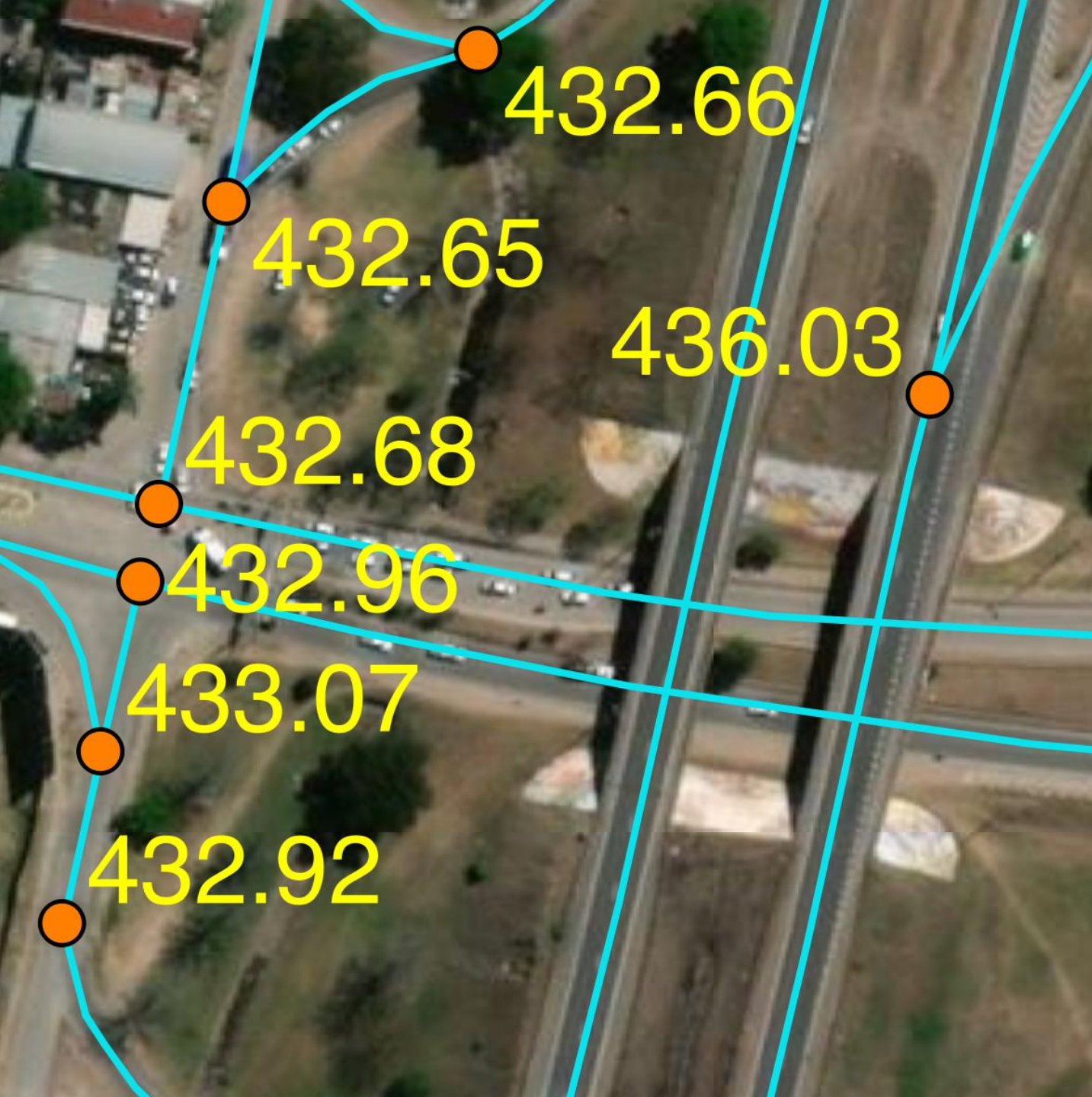




|                         |                           |
|-------------------------|---------------------------|
| Network ID              | 2002                      |
| Closest city            | Nogales                   |
| Country                 | Mexico                    |
| World Region            | Latin America & Caribbean |
| Elevated road structure | Overpass                  |
| Difference in elevation | 6.23 meters               |



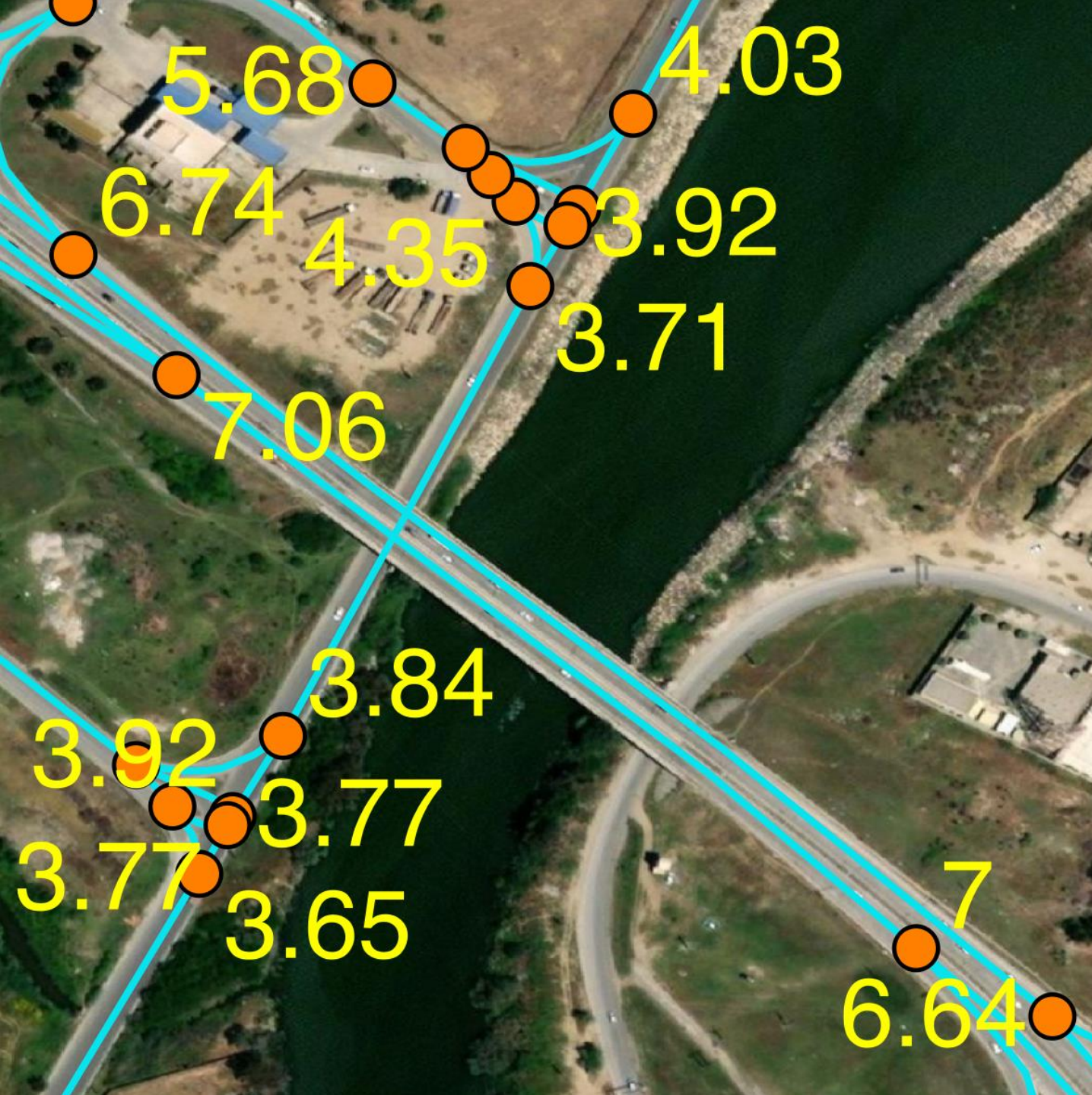




|                         |                           |
|-------------------------|---------------------------|
| Network ID              | 4103                      |
| Closest city            | San Miguel de Tucumán     |
| Country                 | Argentina                 |
| World Region            | Latin America & Caribbean |
| Elevated road structure | Overpass                  |
| Difference in elevation | 3.35 meters               |



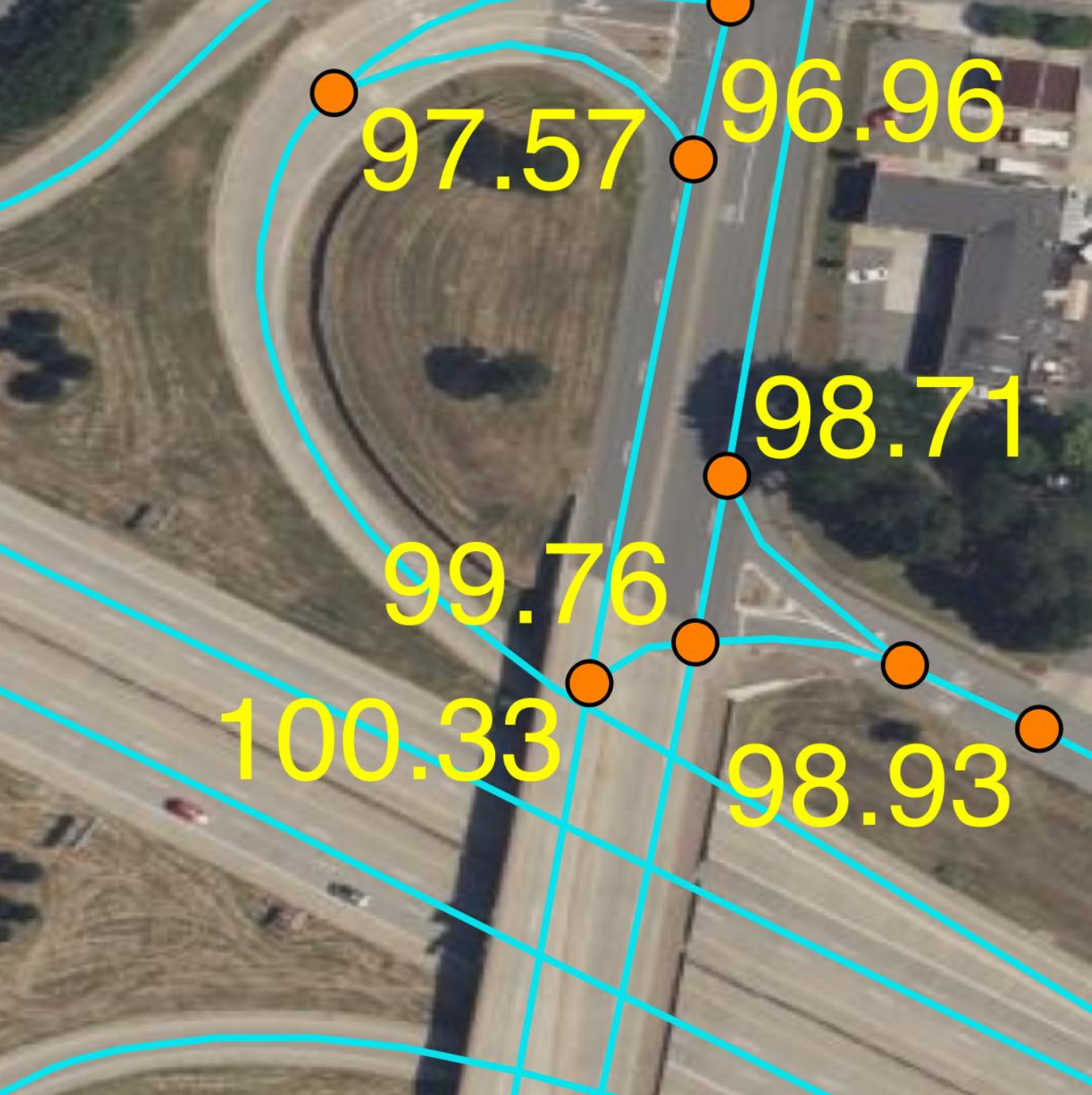




|                         |                            |
|-------------------------|----------------------------|
| Network ID              | 1521                       |
| Closest city            | Annaba                     |
| Country                 | Algeria                    |
| World Region            | Middle East & North Africa |
| Elevated road structure | Bridge                     |
| Difference in elevation | 3.22 meters                |



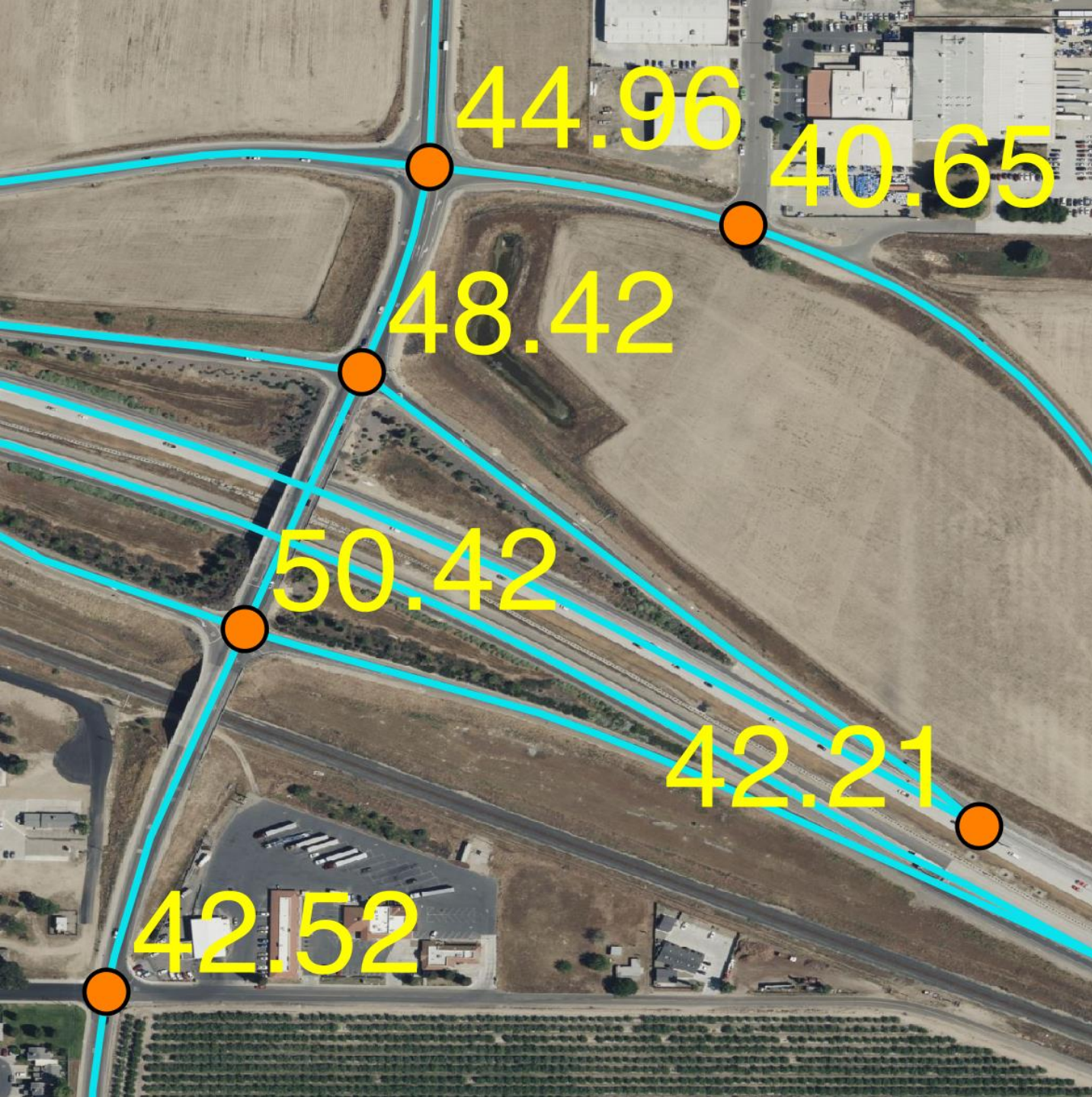




|                         |               |
|-------------------------|---------------|
| Network ID              | 1754          |
| Closest city            | Little Rock   |
| Country                 | United States |
| World Region            | North America |
| Elevated road structure | Overpass      |
| Difference in elevation | 3.37 meters   |







|                         |               |
|-------------------------|---------------|
| Network ID              | 1456          |
| Closest city            | Turlock       |
| Country                 | United States |
| World Region            | North America |
| Elevated road structure | Overpass      |
| Difference in elevation | 8.21 meters   |

